User Interface Design

(Lecture 7)

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Introduction

- In any software product,
  - user interface portion is responsible for all interactions with the user.
- Almost every software product has a user interface:
  - can you think of any software product that does not have any user interface?

Introduction

- In the early days of computer, no software product had any user interface:
  - all computers were batch systems
  - no interactions with users were supported.
- We know that things are very different now:
  - almost every software product is highly interactive.

Introduction

- Users interact with a software product through its user interface:
  - user-interface portion of any software is directly relevant to the users.
  - many users judge a software from its user interface.
- User interface design:
  - a practical and important problem.

Introduction

- Aesthetics apart, a difficult to use interface:
  - leads to higher levels of user errors
  - leads to user dissatisfaction.
- Users become particularly irritated when a system behaves in unexpected ways,
  - issued commands do not carry out actions according to intuitive expectations of users.

Introduction

- A significant portion of the total development effort:
  - spent in developing the user interface.
- For many interactive applications:
  - as much as 50% of the total development effort is spent on the user interface part.

Introduction

- If the user interface is not developed systematically:
  - effort required to develop the interface would increase tremendously :
    - leading to customer dissatisfaction.

Introduction

- It is important to carefully study
  - user interface design concepts.
- Let us examine what a good interface is:
  - Unless we know what a good user interface really is,
    - we cannot possibly design one.

Characteristics of Good User Interfaces

- **Speed of learning:**
  - A good user interface should be simple to learn.
  - Most users are put off by complex syntax and semantics of the command issue procedures.

Speed of learning

- A good user interface should not require
  - users to memorize commands.
- An important factor affecting the speed of learning is consistency:
  - Once, a user learns about a command,
  - should be able to use similar commands
    - even in different circumstances for carrying out similar actions.

Speed of learning

- Users can learn about an interface faster, if it is based on:
  - day-to-day real life examples (aka metaphors)
  - use concepts with which users are already familiar.

Speed of learning

- For example, interface of a text editor:
  - can use concepts similar to writing on paper:
    - such as cutting lines and paragraphs and pasting it at other places,
    - users can immediately relate to it.
- Also, learning is facilitated by:
  - intuitive command names
  - symbolic command issue procedures.

Speed of use

- Speed of use is determined by:
  - the time and effort necessary to initiate and execute different commands.
  - The time and user effort necessary to execute different commands should be minimal.

Speed of use

- **Examples of bad interfaces:**
  - users required to type in lengthy commands
  - command issue involves moving the mouse to widely different areas of the screen
  - can slow down the operating speed of users.

Speed of recall

- Once users learn how to use an interface:
  - their speed of recall about using the interface should be maximized.
- The speed of recall is improved if the interface is based on:
  - metaphors
  - symbolic command issue procedures
  - intuitive command names.

Error rate

- A good user interface should
  - minimize the scope of committing errors.
- Error rate can be easily measured:
  - count the errors committed by different users.

Error rate

- Error monitoring can be automated:
  - instrument user interface with monitoring code
  - record the frequency and types of errors committed by different users
  - later display statistics of various kinds of user errors.

How to reduce error possibilities?

- Consistency of command names,
- Consistency of command issue procedures,
- Consistency in behavior of similar commands
- Simplicity of command issue procedure, etc.

Attractiveness

- A good user interface should be attractive:
  - An attractive user interface catches user attention and fancy.
  - In this respect,
    - graphics-based user interfaces have a definite advantage over text-based interfaces.

Consistency

- **Consistency of commands is very desirable.**
  - allow users to generalize the knowledge about one aspect of the interface to another.
- **Consistency helps in:**
  - speed of learning,
  - speed of recall,
  - also helps in reduction of error rate.

Feedback

- **A good user interface must provide constant feedback to user actions:**
  - For example, if any user request takes more than few seconds to process,
    - the user must be informed that his/her request is still being processed.
  - If possible, the user should be periodically informed about the progress made in processing the command.

Feedback

- In the absence of any response from the computer for a long time:
  - a novice user might even start recovery/shutdown procedures in panic.

Support for multiple skill levels

- **A good user interface:**
  - should support different levels of sophistication in command issue procedures:
  - users with different experience levels prefer different types of interfaces.
- Experienced users are more concerned about speed of command issue:
  - whereas novice users pay prime importance to usability aspects.

Support for multiple skill levels

- Novice users discouraged by:
  - Cryptic/puzzling command names
  - complex command issue procedures.
- Elaborate commands:
  - slow down command issue procedure
  - put off experienced users.

Support for multiple skill levels

- As users become familiar with an interface:
  - look for faster command issue procedures such as " hot-keys", "macros", etc.

Error Recovery (Undo facility)

- All categories of users commit errors.
  - A good interface should allow users to undo mistakes..
- Users are inconvenienced:
  - if they can not recover from even simple errors.

User Guidance and On-line Help

- Users might need guidance
  - or seek help from the system.
- User Guidance is provided through two broad category of methods:
  - On-line help system
  - Guidance and error messages produced
    - in response to user actions.

On-line Help System

- Generic help messages are not very useful:
  - on-line help messages should be tailored to the context in which help is invoked.

On-line Help

- **A good on-line help should:**
  - provide messages in context-dependent way.
    - keep track of what a user is doing
  - help messages should be tailored to user's experience level.
  - should take advantage of graphics capabilities of the screen
    - not just be a copy of the user manual.

Guidance Messages

- The guidance messages should be carefully designed:
  - prompt the user:
    - next actions he/she might take,
    - current status of the system,
    - progress made so far in processing the command

Guidance Messages

- **A good guidance system should have:**
  - different levels of sophistication for different categories of users.
  - users should have an option to turn off detailed messages.

Error Messages

- Error messages should be polite.
- Error messages should not be associated with noise:
  - might embarrass the user.
- The messages should suggest how a given error can be rectified.

Error Messages

- If appropriate,
  - the user should be given the option of invoking on-line help
  - to find out more about the error situation.

Improper Error Message

File not found!

Mode-Based VS. Modeless Interface

- **A mode is a state or collection of states:**
  - in each state (or mode) different types of commands are available.
- **Modeless interface:**
  - same set of commands are available at all times.
- **Mode-based interface:**
  - different sets of commands are available depending on the mode in which the system is, i.e.
  - based on the past sequence of the user commands.

Mode-Based VS. Modeless Interface

- A mode-based interface can be represented using a state transition diagram:
  - each node (state) of the state transition diagram represents a mode.
  - Every state (node) of the diagram
    - annotated with command names meaningful in that state.

Graphical User Interface (GUI) versus Text-Based User Interface

- In a GUI:
  - several windows with different information can be simultaneously displayed on user's screen.
  - This is perhaps the biggest advantage of GUI
  - user can simultaneously interact with several related items at any time
  - can even run many unrelated applications

Graphical User Interface (GUI) versus Text-Based User Interface

- Iconic information representation and symbolic information manipulation is possible in a GUI.
- Symbolic information manipulation:
  - such as pulling an icon representing a file into a trash can for deleting
  - intuitively very appealing
    - user can instantly remember it.

Graphical User Interface (GUI) versus Text-Based User Interface

- A GUI can support command selection:
  - using an attractive and user-friendly menu selection system.
- In a GUI, a pointing device can be used:
  - a mouse or a light pen to issue commands.
  - The use of a pointing device
    - makes command issue procedure simpler.

Graphical User Interface (GUI) versus Text-Based User Interface

- On the flip side, a GUI requires:
  - special terminals with graphics capabilities
  - requires special input devices such a mouse.
- In contrast, a text-based interface:
  - can run even on cheap alphanumeric terminals:
  - Graphics terminals are usually much more expensive than alphanumeric terminals.

Graphical User Interface (GUI) versus Text-Based User Interface

- Increasing availability of:
  - terminals with graphics capability
  - bit-mapped high-resolution displays
  - significant amount of local processing power.
- We will concentrate our attention to GUIs

Types of User Interfaces

- User interfaces can be classified into three categories:
  - Command language-based interface
  - Menu-based interface
  - Direct manipulation interface

Types of User Interfaces

- Each category of interface has its advantages and disadvantages:
  - Modern applications sport a combination of all the three types of interfaces.

Choice of Interface

- Which parts of the interface should be implemented using what type of interface?
  - No simple guidelines available
  - to a large extent depends on the experience and discretion of the designer.
  - a study of characteristics of the different interfaces would give us some idea.

Command Language-Based Interface

- As the name itself suggests:
  - incorporates some language to form commands.
- Users frame the required commands in the language:
  - type them in whenever required.

Design of command language interface

- Simple command language interface:
  - determine all the commands needed to be supported
  - assign unique names to the different commands.

Design of command language interface

- A more sophisticated command language interface:
  - allow users to compose primitive commands to form more complex commands.
    - Consider cat x.dat|grep 123
  - Like a programming language.

Command Language-Based Interface

- **The facility to compose commands:**
  - dramatically reduces the number of command names users would have to remember.
  - Commands can be made concise
    - requiring minimal typing by the user.
  - allow faster interaction with the computer
  - simplify input of complex commands.

Advantages of Command Language Interfaces

- **Easy to develop:**
  - compiler writing techniques are well developed.
- **Can be implemented even on cheap alphanumeric terminals.**
- **Much more efficient:**
  - compared to other types of interfaces.

Disadvantages of Command Language Interfaces

- **Difficult to learn:**
  - Require the user to memorize primitive commands.
- **Require the user to type in commands.**
- **Users make errors while:**
  - formulating commands in the command language
  - typing them in.

Disadvantages of Command Language Interfaces

- All interactions are through key-board:
  - cannot take advantage of effective interaction devices such as a mouse.
  - For casual and inexperienced users,
    - command language interfaces are not suitable.

Issues in Designing a Command Language Interface

- Design of a command language interface:
  - involves several issues.
- The designer has to decide
  - what mnemonics are to be used for the commands.
  - mnemonics should be meaningful
    - yet be concise to minimize the amount of typing required.

Issues in Designing a Command Language Interface

- The designer has to decide:
  - whether users will be allowed to redefine command names to suit their own preferences.
    - Letting a user define his own mnemonics for various commands is a useful feature,
    - but increases complexity of user interface development.

Issues in Designing a Command Language Interface

- Designer has to decide:
  - whether it should be possible to compose primitive commands to create more complex commands.
  - syntax and semantics of command composition options has to be clearly and unambiguously decided.

Issues in Designing a Command Language Interface

- The ability to combine commands is a powerful facility for experienced users:
  - but quite unnecessary for inexperienced users.

Menu-Based Interface

- Advantages of a menu-based interface over a command language interface:
  - users are not required to remember exact command names.
  - typing effort is minimal:
    - menu selections using a pointing device.
    - This factor becomes very important for the occasional users who can not type fast.

Menu-Based Interface

- For experienced users:
  - menu-based interfaces is slower than command language interfaces
  - experienced users can type fast
  - also get speed advantage by composing simple commands into complex commands.

Menu-Based Interface

- Composition of commands in a menu-based interface is not possible.
  - actions involving logical connectives (and, or, all, etc.)
    - awkward to specify in a menu-based system.

Menu-Based Interface

- If the number of choices is large,
  - it is difficult to design a menu-based interface.
  - Even moderate sized software needs hundreds or thousands of menu choices.
- A major problem with the menu-based interface:
  - structuring large number of menu choices into manageable forms.

Structuring Menu Interface

- Any one of the following options is adopted to structure menu items.
  - **Walking menu**
  - **Scrolling menu**
  - **Hierarchical menu**

Scrolling Menu

- Used when the menu options are highly related.
  - For example text height selection in a word processing software.
- Scrolling of menu items
  - lets the user to view and select the menu items that can not be accommodated on one screen.

Walking Menu

- Walking menu is commonly used to structure large menu lists:
  - when a menu item is selected,
  - it causes further menu items to be displayed adjacent to it in a submenu.

Walking Menu

- A walking menu can successfully structure commands only if:
  - there are tens rather than hundreds of choices
  - each adjacently displayed menu does take up some screen space
    - the total screen area is after all limited.

Hierarchical menu

- Menu items are organized in a hierarchy or tree structure.
  - Selecting a menu item causes the current menu display to be replaced by an appropriate submenu.
  - One can consider the menu and its various submenu to form a hierarchical tree-like structure.

Hierarchical menu

- Walking menu are a form of hierarchical menu:
  - practicable when the tree is shallow.
- Hierarchical menu can be used to manage large number of choices,
  - but, users face navigational problems
    - lose track of where they are in the menu tree.

Direct Manipulation Interface

- Present information to the user
  - as visual models or objects.
- Actions are performed on the visual representations of the objects, e.g.
  - pull an icon representing a file into an icon representing a trash box, for deleting the file.
- Direct manipulation interfaces are sometimes called as **iconic interfaces**.

Direct Manipulation (Iconic) Interface

- Important advantages of iconic interfaces:
  - icons can be recognized by users very easily,
  - icons are language-independent.
- However, experienced users consider direct manipulation interfaces too slow.

Direct Manipulation (Iconic) Interface

- It is difficult to form complex commands using a direct manipulation interface.
- For example, if one has to drag a file icon into a trash box icon for deleting a file:
  - to delete all files in a directory one has to perform this operation again and again
  - very easily done in a command language-interface by issuing a command delete *.*

Windowing Systems

- Most modern GUIs are developed using some **windowing system**.
- A windowing system can generate displays through a set of windows.
- Since a window is a basic entity in such a graphical user interface:
  - we need to first discuss what exactly a window is.

Window

- **A window is a rectangular area on the screen.**
- **A window is a is a virtual screen:**
  - it provides an interface to the user for carrying out independent activities,
  - one window can be used for editing a program and another for drawing pictures, etc.
- **A window can be divided into two parts:**
  - **client part,**
  - **non-client part.**

Window

- The client area makes up the whole of the window,
  - except for the borders and scroll bars.
- The client area is available to the programmer.
- Non-client area:
  - under the control of window manager.

Window management system (WMS)

- A graphical interface might consist of a large number of windows,
  - necessary to have some systematic way to manage the windows.
  - Window Management System (WMS)

Window management system (WMS)

- GUIs are developed using a window management system (WMS):
  - **A window management system is primarily a resource manager.**
  - keeps track of screen area resource
  - allocates it to the different windows which are using the screen.

Window management system (WMS)

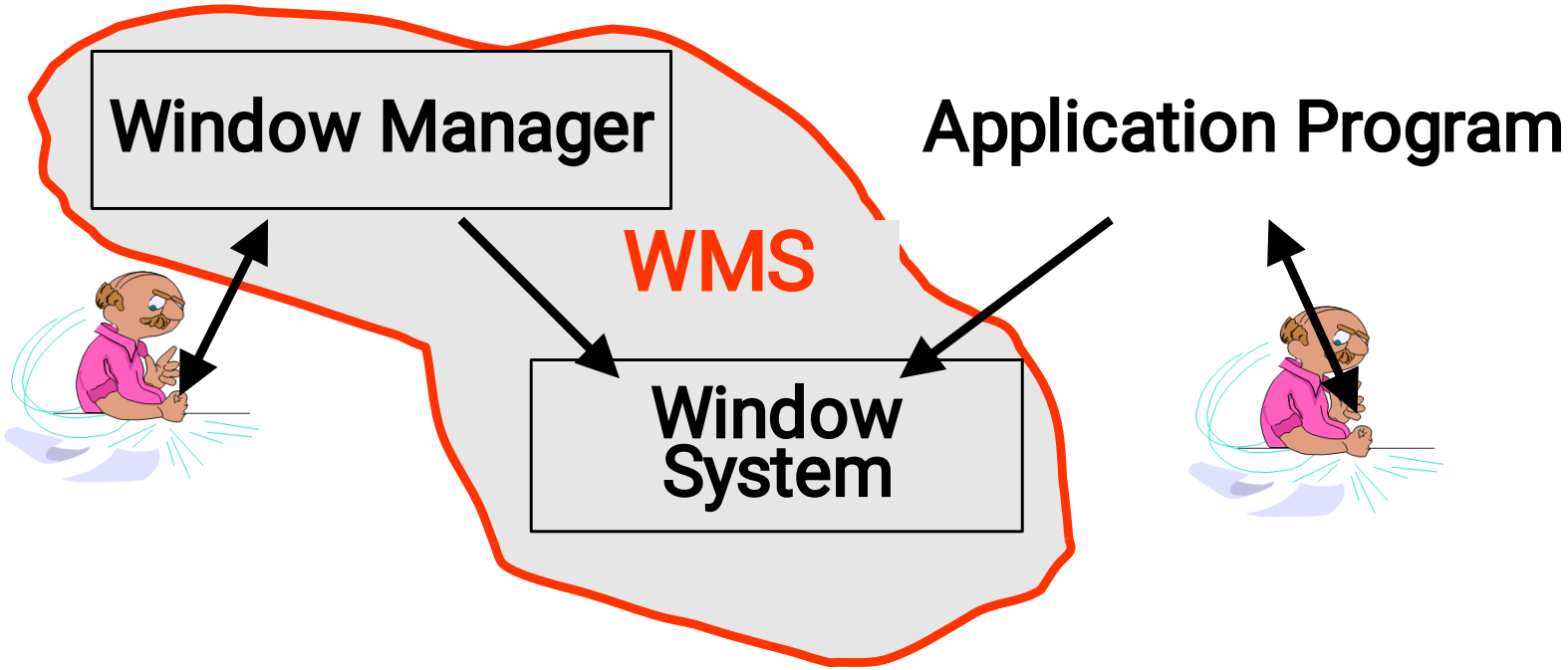
- From a broader perspective, a WMS can be considered as a **user interface management system (UIMS) ---**
  - not only does resource management,
  - also provides the basic behavior to windows
  - provides several utility routines to the application programmer for user interface development.

Window management system (WMS)

- **A WMS simplifies the task of a GUI designer to a great extent:**
  - provides the basic behavior to the various windows such as move, resize, iconify, etc.
  - **provides routines to manipulate windows such as:**
    - creating, destroying, changing attributes of the windows, and drawing text, lines, etc.

Window management system (WMS)

- A WMS consists of two part:
  - a window manager
  - a window system.



Window Manager and Window System

- User interacts with window manager to do various window-related operations such as:
  - **window repositioning,**
  - **window resizing,**
  - **iconification, etc.**
- Window manager also controls the screen's real estate policy.

Window manager

- The window manager is built on the top of the window system:
  - makes use of the basic services provided by the window system.
- The window manager determines how the windows look and behave.
  - several kinds of window managers can be based on the same window system.

Window manager

- Window manager can be considered as a special program:
  - makes use of the services (function calls) of the window system.
- Application programs
  - invoke the window system for user interface-related functions.

Window System

- Provides a large number of routines for the programmer
- It is very cumbersome to use these large set of routines:
  - most WMS provide a higher-level abstraction called **widgets.**

Window Management System

- A widget is the short form for a window object.
- Widgets are the building blocks in interface design.
- We know that an object is essentially a collection of:
  - related data with several operations defined on these data.

Widgets

- The data of an window object are:
  - the geometric attributes (such as size, location etc.)
  - other attributes such as its background and foreground color, etc.
- The operations defined on these data include, resize, move, draw, etc.

Advantages of Widgets

- One of the most important reasons to use widgets as building blocks:
  - provide consistency.
- Consistent user interfaces
  - improve the user's productivity and
  - lead to higher performance with fewer errors.

Advantages of Widgets

- Widgets make users familiar with standard ways of using an interface ---
  - users can easily extend their knowledge of interface of one application to another
  - the learning time for users is reduced to a great extent.
- Next lecture we will identify standard widgets used to design GUIs.

Summary

- User interface is an important component of any software product.
- We first discussed some important concepts associated with user interface design.
- We discussed some desirable properties of a good user interface.

Summary

- There are 3 main types of interfaces:
  - **command language interface**
  - **menu-based interface**
  - **iconic interface**
- Each type of interface has its own advantages and disadvantages:
  - most modern interfaces use a combination of all the three types.

Summary

- We finally discussed concepts associated with a window management system (WMS):
  - consists of a window manager and a window system.
  - we discussed the use of widgets as the basic building blocks in GUI design.

Thank you!